ECO-CONNECT

Reimagining Telkomsel's Consumer Touchpoints Through Agentic Multimodal AI, Green Monetization, and Next-Gen Experience Hubs

Simulated Business Case Solution for Telkomsel – International Business Case Competition (IBCC) 2026
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1. Problem Statement: Decoupling Traffic Surges from Flat ARPU & Carbon Drag

+25% vs +3%

MONETIZATION SQUEEZE

Data traffic expands +25%/yr while blended ARPU crawls from Rp45k to Rp49k, squeezing EBITDA margins.

15M Visits/Yr

GRAPARI COST DRAIN

±350 outlets handle 15M routine visits costing ~Rp375B annually, with 60% technically eligible for self-service.

37,500 tCO_2e

CARBON FOOTPRINT

Customer travel (10–15 km roundtrip) generates 2.5 kg CO_2e per physical visit, hindering 2030 net-zero goals.

Root Cause Diagnosis (Why 60% of Visits Remain Physical):

1. Digital Literacy Friction: Elderly & rural segments struggle with complex app navigation and English terms.

2. Biometric Trust Deficit: Lost SIM & e-KYC swap require physical KTP photocopying and in-person verification.

3. Absence of Green Incentives: Customers lack tangible gamified motivation to choose digital channels over walk-in.

2. Solution Idea: The Three Pillars of ECO-CONNECT

Pillar 1: Agentic Veronika 2.0 & Inclusive e-KYC

- **Vernacular Voice AI:** Voice commands in Bahasa, Javanese & Sundanese via Azure OpenAI.
- **WhatsApp-Native Self-Service:** 100% flow execution without forcing app downloads.
- **Biometric Remote SIM Swap:** Dukcapil liveness verification for instant eSIM / courier delivery.

🎯 Impact: Shifts 7.5M visits to digital by Year 3 (SDGs 9 & 10).

Pillar 2: Green Eco-Pass & Monetized VAS

- **Telkomsel Eco-Pass:** In-app tracker rewarding 2.47 kg CO_2e saved per digital shift with Eco-Poin.
- **Monetized AI VAS:** "Veronika CyberGuard" (SMS scam filter, banking protector) @ Rp10,000/mo.
- **B2B ESG Syndication:** Anonymized mobility carbon data sold to enterprise clients.

🎯 Impact: +Rp306B new revenue & +Rp3,000 ARPU by Y3 (SDGs 12 & 13).

Pillar 3: GraPARI NextGen Experience Hubs

- **24/7 Smart Exterior Kiosks:** Automated physical SIM dispensing, cutting counter space by 60%.
- **MSME Digitalization Lounges:** Interior repurposed for 5G, Smart Home (IndiHome), & SME labs.
- **High-Touch Concierge:** Staff upskilled for complex disputes, fraud, and senior guidance.

🎯 Impact: Saves Rp172.5B OpEx/yr; transforms cost drain into retail asset.

3. Technical & Operational Feasibility: Solution Architecture



1. Touchpoint Integration: Omnichannel integration across MyTelkomsel (50M MAU), WhatsApp Business API, and 24/7 Smart Kiosks.

2. Azure OpenAI Orchestration: Intent classification, autonomous tool execution via Telkomsel Core BSS/OSS APIs, and local SLM voice models.

3. Dukcapil Security Fabric: Encrypted telco SIM-IMEI binding with government biometric liveness check to prevent fraud and deepfakes.

4. Business & Financial Viability: High-Margin Dual-Engine Returns

Financial Metric (IDR Billion)	Y1 (2027)	Y2 (2028)	Y3 (2029)
Shifted Physical Visits (M)	2.7M	5.0M	7.5M
GraPARI OpEx Savings (@Rp23k)	Rp 62.1	Rp 115.0	Rp 172.5
New AI VAS (CyberGuard @Rp10k)	Rp 96.0	Rp 180.0	Rp 264.0
Eco-Pass Brand Partnerships	Rp 15.0	Rp 28.0	Rp 42.0
Total Gross Economic Benefit	Rp 173.1	Rp 323.0	Rp 478.5
Total CapEx & OpEx Investment	(Rp 67.0)	(Rp 40.0)	(Rp 32.0)
Net Cash Flow (NCF)	Rp 106.1	Rp 283.0	Rp 446.5

Rp 672.4 B

NET PRESENT VALUE (10% WACC)

Exceptional financial return with **142% IRR** and an aggressive **7.8-Month Payback Period**.

37,544 Tons CO_2e

CUMULATIVE CARBON AVOIDED (GHG SCOPE 3)

7.5M shifted visits eliminate 2.47 kg CO_2e each, equivalent to planting **620,000 mature trees** and saving **6.25M hours**.

5. Collaboration & Implementation Fit: Seamless Organizational Alignment

Internal Cross-Functional Governance

Operated via dedicated Agile Squads on 2-week sprint cadences:

- **Digital Touchpoint Squad:** Drives MyTelkomsel UI/UX & WhatsApp conversational flows.
- **AI & Network Engineering:** Fine-tunes Azure OpenAI models & integrates Core BSS/OSS APIs.
- **Retail Operations & GraPARI Squad:** Coordinates smart kiosk retrofitting and staff upskilling.
- **ESG Directorate:** Audits GHG Scope 3 emission accounting and carbon credit registries.

External Strategic Ecosystem

Key multi-stakeholder partnerships:

- **Directorate General of Dukcapil:** Real-time national biometric e-KYC integration.
- **Microsoft Azure AI Team:** Enterprise-grade OpenAI infrastructure & zero-data-retention security.
- **Green Merchant Ecosystem:** Electric mobility (EV charging), mass transit, & eco-merchants for Eco-Poin redemption.
- **GraPARI Workforce Compact:** Zero-layoff pledge; staff re-skilled as MSME Digital Transformation Advisors.

6. Execution Readiness & Phased 36-Month Roadmap

Horizon 1: Pilot & Validate (M1–M6)

- Fine-tune vernacular AI in Javanese & Sundanese.
- Integrate Dukcapil biometric liveness API.
- Pilot WhatsApp SIM swap & 30 smart kiosks in Jabodetabek.
- **KPI:** 85% first-contact resolution; 300k visits shifted.

Horizon 2: Scale Nationwide (M7–M18)

- Deploy 350+ 24/7 Smart Kiosks across all branches.
- Launch nationwide Eco-Pass gamification campaign.
- Commercialize Veronika CyberGuard VAS (@Rp10k/mo).
- **KPI:** 5.0M visits shifted; +Rp180B VAS revenue.

Horizon 3: Ecosystem Maturity (M19–M36)

- Complete GraPARI transition into MSME Hubs.
- Syndicate B2B Scope 3 carbon credits to corporate clients.
- Scale AI Concierge to full FMC home automation.
- **KPI:** 7.5M visits shifted; 37.5k tCO_2e avoided.

Critical Risk Governance: Biometric Fraud: Multi-factor liveness + telco SIM lock | **Elderly Friction:** GraPARI Digital Concierge staff on floor | **AI Hallucination:** Deterministic guardrails with instant fallback to human contact center < 30s.

7. Closing & Support Needed: Building Indonesia's Green Telco Champion

The ECO-CONNECT Value Dividend

- **Decouples Growth from Emissions:** Eliminates 37,544 tCO_2e, directly contributing to Telkomsel's 2030 -20% carbon intensity mandate.
- **Closes the Monetization Gap:** Delivers Rp835.6B cumulative net benefit and lifts ARPU by +Rp3,000 for active digital users.
- **Drives True Digital Inclusion:** Empowers non-digital natives via local dialects and zero-download WhatsApp self-service (SDGs 9, 10).
- **Future-Proofs Physical Assets:** Reinvents 350 GraPARI outlets from costly complaint desks into high-margin experience hubs.

Support & Resources Required from Telkomsel

- **Executive Sponsorship:** Joint mandate from Director of Marketing and Director of Network & IT.
- **Regulatory Facilitation:** Priority coordination with Dukcapil for high-throughput biometric API access.
- **CapEx Budget Allocation:** Rp42 Billion Year 1 investment for kiosk procurement and voice AI fine-tuning.
- **Retail Operations Alignment:** Phased transition plan for GraPARI customer service workforce upskilling.

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  "Together, let's empower every Indonesian home and business sustainably."
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